BRITISH STANDARD

BS EN 50122-2:1999

Incorporating Corrigendum No. 1 and Amendment No. 1

Railway applications — Fixed installations —

Part 2: Protective provisions against the effects of stray currents caused by d.c. traction systems

The European Standard EN 50122-2:1998, with the incorporation of amendment A1:2002, has the status of a British Standard

ICS 29.120.50; 29.280

British Standards

National foreword

This British Standard is the English language version of EN 50122-2:1998, including Corrigendum August 2001 and amendment A1:2002.

The UK participation in its preparation was entrusted by Technical Committee GEL/9, Railway electrotechnical applications, to Subcommittee GEL/9/3, Fixed equipment, which has the responsibility to:

— aid enquirers to understand the text;
— present to the responsible European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;
— monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

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(includes amendment A1:2002)

Applications ferroviaires —
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(inclut l'amendement A1:2002)

Bahnanwendungen — Ortsfeste Anlagen —
Teil 2: Schutzmaßnahmen gegen die Auswirkungen von Streuströmen verursacht durch Gleichstrombahnen
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CENELEC

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Europäisches Komitee für Elektrotechnische Normung

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Foreword

This European Standard was prepared by SC 9XC, Electric supply and earthing systems for public transport equipment and ancillary apparatus (fixed installations), of Technical Committee CENELEC TC 9X, Electrical and electronic applications for railways.

The text of the draft was submitted to the formal vote and was approved by CENELEC as EN 50122-2 on 1998-04-01.

The following dates were fixed:

— latest date by which the EN has to be implemented at national level by publication of an identical national standard or by endorsement	(dop) 1999-03-01
— latest date by which the national standards conflicting with the EN have to be withdrawn	(dow) 1999-03-01

Foreword to amendment A1

This amendment to the European Standard was prepared by SC 9XC, Electric supply and earthing systems for public transport equipment and ancillary apparatus (fixed installations), of Technical Committee CENELEC TC 9X, Electrical and electronic applications for railways.

The text of the draft was submitted to the Unique Acceptance Procedure and was approved by CENELEC as amendment A1 to EN 50122-2:1998 on 2002-04-01.

The following dates were fixed:

— latest date by which the amendment has to be implemented at national level by publication of an identical national standard or by endorsement	(dop) 2003-04-01
— latest date by which the national standards conflicting with the amendment have to be withdrawn	(dow) 2005-04-01

Contents

1 Scope

This standard specifies requirements for protective provisions against the effects of stray currents which result from the operation of d.c. traction systems.

As experience for several decades has not shown evident corrosion effects from a.c. traction systems and actual investigations are not completed, this standard only deals with stray currents flowing from a d.c. traction system.

This standard applies to all metallic fixed installations which form part of the traction system, and also to any other metallic components located in any position in the earth, which may carry stray currents resulting from the operation of the railway system.

This standard applies to all new electrification of a d.c. railway system. The principles may also be applied to existing electrified systems where it is necessary to consider the effects of stray currents.

The range of application includes:

— railways;

— guided mass transport systems such as:

tramways, elevated and underground railways, mountain railways, trolleybus systems and magnetic levitated systems;

— material transportation systems.

This standard does not apply to:

a) mine traction systems in underground mines;

b) cranes, transportable platforms and similar transportation equipment on rails, temporary structures (e.g. exhibition structures) in so far as these are not supplied directly from the contact line system and are not endangered by the traction power supply system;

c) suspended cable cars;

d) funicular railways;

e) maintenance work.

2 Normative references

This European Standard incorporates by dated or undated references, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

EN 12954[1)], *Cathodic protections of buried or immersed metallic structures — General principles.*

EN 50122-1, *Railway applications — Fixed installations — Part 1: Protective provisions relating to electrical safety and earthing.*

IEC 60050(826), *International Electrotechnical Vocabulary — Chapter 826: Electrical installations of buildings.*

3 Definitions

For the purposes of this standard, the following definitions apply.

3.1
stray current
a current which follows paths other than the intended paths

3.2
stray current zone
zone in which currents may be exchanged between a d.c. traction system and metallic structures or earth

NOTE Such a stray current zone may extend over a distance of a few kilometres.

[1)] In preparation.

3.11
return cable busbar
a busbar in substations at which return cables terminate

3.12
d.c. (traction) substation
an installation the main function of which is to supply a contact line system, at which the voltage of a primary supply system, is converted to the voltage of the contact line

3.13
traction return current
the sum of the traction currents returning to the supply source

NOTE The supply source may be a regenerating vehicle.

3.14
rail joint bond
a conductor ensuring the electrical continuity of a rail at a joint

3.15
insulated rail joint
a mechanical rail joint which longitudinally separates the rail electrically

3.16
rail-to-rail cross bond
an electrical bond that interconnects the running rails of the same track

3.17
track-to-track cross bond
an electrical bond that interconnects tracks

3.18
earth
the conductive mass of the earth, whose electric potential at any point is conventionally taken as zero [IEC 50, **826-04-01**]

3.19
earth electrode
a conductive part or a group of conductive parts in intimate contact with and providing an electrical connection with earth [IEC 50, **826-04-02**]

3.20
tunnel earth
the electrical interconnection of the reinforcing rods of reinforced concrete tunnels, and in the case of other modes of construction, the conductive interconnection of the metallic parts of the tunnel

3.21
structure earth
the electrical interconnection of the reinforcing rods of structures, and in the case of other modes of construction, the conductive interconnection of the metallic parts. Examples are reinforced railway structures such as bridges, viaducts and reinforced trackbed

3.22
rail to earth resistance
the electrical resistance between the running rails and the earth

NOTE 1 For d.c. traction systems in tunnels the measurement is made between the running rails and tunnel earth.

NOTE 2 In this standard the rail to earth resistance refers to a single-track installation, unless otherwise explicitly specified.

3.23
conductance per unit length
the reciprocal value of the rail to earth resistance per unit of length

NOTE In this standard the conductance per unit length refers to a single-track installation, unless otherwise explicitly specified.

3.24
equipotential bonding
electrical connection putting various exposed conductive parts and extraneous conductive parts at a substantially equal potential [IEC 50, **826-04-09**]

3.25
equipotential bonding conductor
a protective conductor for ensuring equipotential bonding [IEC 50, **826-04-10**]

3.26
rail potential
the voltage occuring under operating conditions when the running rails are utilized for carrying the traction return current, or under fault conditions, between running rails and earth

3.27
(effective) touch voltage
voltage under fault conditions between parts when touched simultaneously

NOTE The value of the effective touch voltage may be appreciably influenced by the impedance of the person in contact with these parts.

3.28
accessible voltage
that part of the rail potential under operating conditions which can be bridged by persons, the conductive path being conventionally from hand to both feet through the body or from hand to hand (horizontal distance of 1 m to a touchable part)

3.29
closed formation
area where the top of the running rails is at the same level as the surface

3.30
open formation
area where the running rails are laid above the surface

3.31
railway authority
the person or organization who owns or is responsible for the overall management of the relevant railway infrastructure

3.32
validation
the process of demonstrating by test and analysis that the system under consideration meets in all respects the specification for that system

3.33
verification
the process of determining for each phase of the safety lifecycle that the output meets in all respects the objectives and requirements set for the specific phase, for example, forward traceability from the requirements specification through each stage of the design documentation to the final design

NOTE Verification may include testing.

3.34
earthing
the connection of conductive parts to an appropriate earth electrode

3.35
open traction system earthing
the connection of conductive parts to the track return system or the track return system to earth by a voltage-limiting device or by circuit-breakers, which make a conductive connection either temporarily or permanently if the limited value of the voltage is exceeded

4 General

4.1 D.C. traction systems may cause stray currents which could adversely affect both the railway concerned and/or outside interests. In order to determine the extent of the problem an assessment study shall be carried out in co-operation with the affected parties. The outcome of this action will assist at the design stage to establish the optimum range of solutions for any identifiable effects. Any provisions adopted to control the effects of stray currents shall be subjected to validation and verification according to the provisions of this standard. If no specific effect/remedy is determined consideration shall be given to the establishment of a regime of periodical checks.

All connections to the track return systems shall be approved by the railway authority.

Protective provisions against electric shock shall take precedence over provisions against the effects of stray currents. See EN 50122-1.

NOTE The major effects of stray currents can be:

— corrosion and subsequent damage of metallic structure where stray currents leave the metallic structures;

— the risk of overheating, arcing and fire and subsequent danger to equipment and persons not necessarily within the railway authority's area of responsibility;

— influence on non-immunized signalling and communication systems;

— influence on unrelated cathodic protection installations;

— influence on unrelated a.c. and d.c. power supply systems.

4.2 The following systems which may produce stray currents shall be considered:

— d.c. traction systems using running rails carrying the traction return current including track sections of other traction systems bonded to the tracks of d.c. traction systems;

— d.c. trolleybus systems which share the same power supply with a system using the running rails carrying the traction return current;

— d.c. traction systems not using running rails carrying the traction return current.

4.3 All systems which may be affected by stray currents shall be considered, such as:

— pipework;

— cables with armour or metal shield;

— tanks and vessels;

— earthing systems;

— concrete constructions containing metal;

— buried metallic structures;

— unrelated cathodic protection installations;

— signalling and telecommunication systems.

5 Traction power supply system

5.1 The traction power supply system, the return circuit and the earthing system shall be considered by a study. In order to allow the assessment of the stray current effects, the study shall include aspects such as:

— distance between the substations;

— return circuit bonding;

— insulation of the rails and other structures from earth;

— additional corrective provisions (see clauses **6** and **7**).

5.2 Where required the output voltage of individual substations shall be adjusted in order to minimize stray current effects.

5.3 If trolleybuses and tramways receive their traction current from the same substation, one of the trolley contact wires may be connected with the track return system at one or several points provided that the track return system allows the continuous passage of current. In this case it shall be checked to determine whether the protective provisions for tramways to minimize stray current effects are still sufficient.

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5.4 The return cable busbars in substations and similar installations shall be operated so that they are insulated from earth. Where required for safety reasons an automatic voltage-limiting device to connect between the return cable busbar and earth shall be provided in accordance with **7.2.6**. For substations in depots and workshops see **7.4**.

5.5 Return cables shall have an insulating outer sheath. Return conductors and return conductor rails shall be insulated from earth.

NOTE Where mechanical damage is likely, return cables should have additional protection.

6 Track system

6.1 Rail system

NOTE 1 In order to minimize stray current caused by a d.c. traction system, the most effective provisions are those which aim at confining the traction return current inside their intended metallic return circuit. This can be achieved either by providing a dedicated conductor for the traction return current (fourth rail), or, when the running rails are used as part of the return circuit, by assuring a high level of insulation of the running rails and of the whole return circuit from earth. Additional improvements can be attained by decreasing the longitudinal resistance of the return circuit.

NOTE 2 When pre-existing earth current distribution is a concern, a high level of insulation of the running rails from earth is also important in limiting the disturbance of the pre-existing earth current distribution by the new railway structures.

6.1.1 *Rail to earth insulation*

6.1.1.1 *Values for conductance per unit length*

For areas where there is a risk of serious stray current effects the conductance per unit length shall be sufficiently low during construction in order that the values given in Table 1 can be maintained during normal operation.

Table 1 — Recommended conductance per unit length *G′* for single track sections

Traction system	Open air S/km	Tunnel S/km
Railway	0,5	0,5
Mass transportation system in open formation	0,5	0,1
Mass transportation system in closed formation	2,5	—

NOTE 1 The values given in Table 1 are based on two running rails per track.

NOTE 2 Provisions to achieve the values for railways and mass transportation systems in open and close formation are:

- — clean ballast;
- — wooden sleepers or concrete sleepers with insulated fastening systems;
- — adequate clearance between running rails and ballast;
- — effective water drainage.

NOTE 3 Provisions to achieve the improved (lower) values for closed formation systems are for instance:

- — embedding the running rails in an insulating resin bed;
- — interposing insulating layers between the tracks and the supporting structure.

NOTE 4 In order to demonstrate compliance with Table 1 a recognized measurement method should be used such as given in Annex A.

6.1.1.2 *Level crossings*

At level crossings, where the running rails are laid in a closed formation, care shall be taken to avoid increasing the conductance per unit length significantly above the typical value of the adjoining track.

6.1.1.3 *Grassed areas*

For tracks in grassed areas special provisions shall be made to achieve and maintain the level of insulation (according to Table 1).

6.1.1.4 *Fourth rail systems*

In the case of return conductor rail (fourth rail), both conductor rails shall be effectively insulated from earth at a level commensurate with the maximum voltages to earth that can be applied to the rails.

6.1.1.5 *Stray current assessments*

When stray current assessments are to be made over a period of time, a recognized method shall be used. Annex A shows a method of determining the value of conductance per unit length.

NOTE Successive measurements should use the same method as used for the first value or the reference value.

6.1.2 *Return circuit resistance*

The longitudinal resistance of the track return system shall be low; therefore rails shall be welded or connected by rail joint bonds of low resistance such that the overall longitudinal resistance of the rails is not increased by more than 5 %.

NOTE The longitudinal resistance can be reduced by:

— use of running rails with greater square section;

— cross bonding of the running rails and/or the tracks (where signalling considerations allow);

— additional cables connected in parallel with the tracks (where signalling considerations allow).

6.2 Other parts of the track system

6.2.1 No part of the traction return circuit shall have direct conductive connections to installations, components or structures which are not insulated from earth.

If the connection with a return circuit is unavoidable for reasons of protection against electric shock, provisions shall be made to reduce the stray current effects. These can be for example:

— open traction system earthing. In this case the voltage-limiting device shall conform to the requirements given in **7.2.6**;

— insulation of the equipment or components that are connected to the running rails, relative to foundations or components that are earthed;

— insulation of the structure from earth.

6.2.2 Insulation of the running rails shall be co-ordinated with other provisions to ensure that accessible voltages due to the traction return currents and touch voltages due to currents under fault conditions do not exceed the admissible values given in EN 50122-1.

6.2.3 Stationary and portable electrical installations and equipment which are connected to the return circuit shall not be supplied directly from a public low-voltage network of the TN system. (Solutions see EN 50122-1, **6.2.4.3**).

6.2.4 The tracks of d.c. traction systems shall not have any direct conductive connection to track sections of other traction systems.

NOTE Track sections of other traction systems can, if necessary, be connected in special cases to the return circuit if they fulfil the requirements given in **6.1**.

However at certain interfaces, particularly those between a.c. and d.c. traction systems, additional provisions shall be made in order to minimize stray current from the common track that forms part of the return circuit for both systems.

Any additional provisions shall not affect other safety criteria particularly accessible voltage limits, and the operation of power supply protection, track circuits and communication systems.

6.2.5 Rail-to-rail cross bonds, track-to-track cross bonds and other bonds which may come in contact with earth shall be insulated from earth. This does not apply to rail tie bars.

6.2.6 Where d.c. traction systems approach buried pipes or cables, efforts shall be made to ensure that the metal parts are kept as far away as practicable to avoid stray currents.

NOTE A minimum distance of 1 m has been found to be adequate for this purpose.

7 Influenced structures

7.1 General

The resistance between conductive structures which are not insulated from earth and the track return system shall be high. Especially a direct connection is not allowed except in the case of depots and workshops (see **7.4**). A direct connection to earth may also be allowed in certain industrial systems with d.c. traction taking into consideration the particular surrounding conditions.

7.2 Tunnel structures

7.2.1 In a tunnel structure which incorporates conductive components it may be necessary to make provisions to limit possible effects of stray currents. The requirements for protection against electric shock shall be taken into account.

NOTE The provisions to reduce the stray current effects in tunnel structures with conductive components can depend on:

— whether the predominant source of the stray current is internal or external to the tunnel;

— whether the main priority is to protect the tunnel metallic structures, or to protect other metallic structures external to the tunnel and the railway.

7.2.2 At the end of tunnels where the tracks outside the tunnel are bedded in a closed formation and when they are not insulated from earth, there shall be installed insulated rail joints in each running rail. The insulated rail joints shall be run open to reduce stray current both into and within the tunnel. In this case a separate tunnel traction power supply is necessary and the overhead contact line of the tunnel shall be electrically segregated from that outside the tunnel.

7.2.3 In the case of tunnels with reinforced concrete structures or other conductive structures it is possible that stray currents can flow into such structures and from there cause influence to other conductive structures outside the tunnel. In this case the effect of such influence shall be reduced by means of equipotential bonding in the lower part of the individual tunnel sections or other conductive structures. For this purpose a sufficient number of reinforcing bars, mats connected together, other conductive structural parts and if necessary, additional conductors of appropriate cross-section laid within the tunnel shall be used.

NOTE 1 As a direct measurement of stray currents is impractical, the potential of the structures against earth is taken for the assessment. Experience has shown that there is no cause for concern, if the average value of the potential shift in the hour of highest traffic does not exceed +100 mV.

In order to avoid inadmissible stray current effects at the tunnel structure and at structures outside of the tunnel, the longitudinal voltage between any two points of the tunnel structure should be calculated. As an example for calculation see Annex C. This is a conservative procedure which ensures that the actual values for the tunnel potential against earth will be lower.

NOTE 2 Individual tunnel sections in particular cases may be excepted from the equipotential bonding of the rest of the tunnel. The equipotential bonding of the other tunnel sections may be achieved by means of an insulated cable extending over the segregated tunnel section.

NOTE 3 For stray current purposes only, it is possible to achieve adequate electrical conductivity of reinforcing bars by means of conventional steel wire wrapping.

7.2.4 In areas where the influence of stray currents on structures outside the tunnel is not significant and when a sufficiently high value of rail to earth resistance cannot be achieved (due to humidity or ballast which is not sufficiently clean) the main consideration shall be the corrosion of the tunnel metallic structures.

Reinforced concrete tunnel structures shall be divided into longitudinal sections by insulated joints in the case where stray currents from other adjacent systems can flow along the tunnel structure, thus causing an undesirable electrical connection between different and very distant city areas.

NOTE 1 If the resistance between these structures and earth is relatively high, for example in rock tunnels, the reinforced concrete tunnel structures may also be divided into longitudinal sections by insulated joints.

If there is any risk of an inadmissible voltage between simultaneously accessible parts, refer to EN 50122-1.

At ring joints between each section terminals shall be provided for test purposes. A reliable electrical connection shall be made between these terminals and the longitudinal reinforcing bars.

NOTE 2 Normally, no connection will be made between the terminals of adjacent sections.

7.2.5 The reinforcement of steel-reinforced concrete tunnels and tunnel components in iron materials shall not have any conductive connection to pipes and cables located outside the tunnel or to the return circuit or to any adjacent systems which are not insulated from earth. Connections of the tunnel reinforcement to its own earth leads in order to satisfy earthing requirements for protective provisions is permissible.

7.2.6 If the voltage-limiting devices between the return circuit and the metallic components of the tunnel structure are provided as a protective provision to prevent inadmissible voltages in accordance with EN 50122-1, the following conditions shall be satisfied.

If the voltage-limiting device has operated, either it shall automatically reset after a maximum of 10 s or, if it is not reset, a procedure shall be established to note and rectify the cause of such events rapidly.

The voltage-limiting device shall be designed to operate under the highest envisaged value of current under fault conditions. Once closed, it shall not open until this current has reduced to a safe value (lower than the rated breaking current of the device).

7.2.7 In tunnel structures which have conductive connections to adjacent buildings (e.g. railway stations, shops) consideration shall be given to monitoring the potential of the track return system of the d.c. traction system so that appearance of low-resistance conductive connections between the return circuit and the structure can be detected. Such connections shall be removed without delay.

7.3 Bridges, viaducts and reinforced trackbed

For such structures the same principles shall be applied as for tunnels.

7.4 Depots and workshops

If in deviation from the general statement in **7.1** a direct connection between the conductive structure and the return circuit is appropriate provisions shall be made as follows.

The running rails in depots or workshops shall be separated from the main line by insulated rail joints and the traction power shall be supplied by separate rectifiers or other means (see EN 50122-1).

7.5 Cables, pipework and power supply from outside

At the entrance of reinforced concrete or metallic railway structures (such as viaducts, depots and workshops), all metallic pipework, hydraulic lines, cable sheaths (either power or telecommunication cables) and connections to earth (e.g. for protective purposes) coming from outside shall be separated electrically from structure in order to avoid any conductive connection between structure earth and external earth electrodes. Metal pipework inside the tunnel shall not bypass insulated ring joints.

NOTE 1 This may be achieved by:

— fitting of insulation parts in the pipes or alternatively complete insulation from the structure earth;

— installation of transformers with separate windings or by the use of the TT or IT systems in accordance with EN 50122-1.

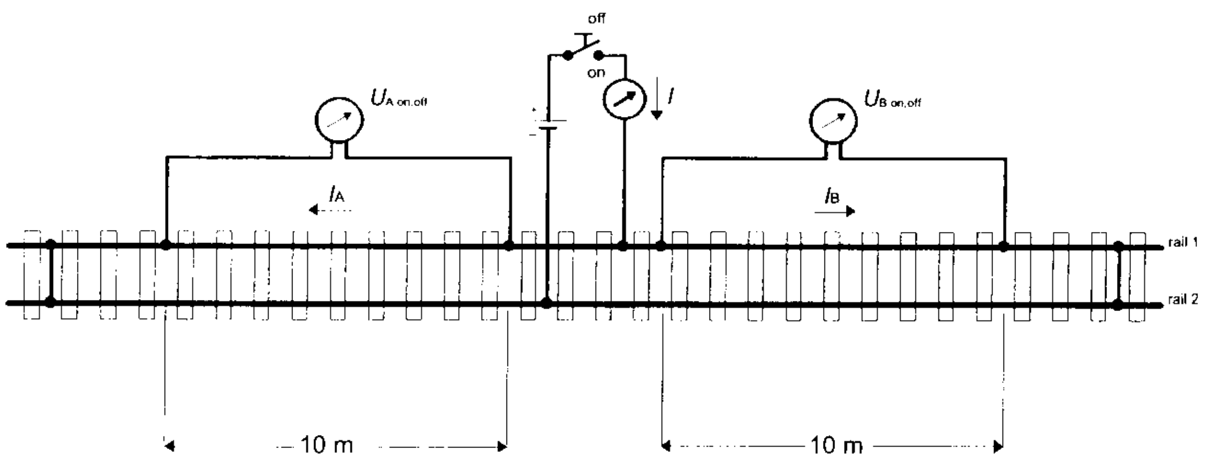
NOTE 2 When necessary for safety reasons, every section of metal pipe may be connected to the metallic structure.

8 Protection methods applied to metallic structures

The provisions in this standard are intended to reduce stray currents and their corrosive effects. Conventional protection methods against natural corrosion can be used if they are considered to be necessary. If additional protection methods are taken into account, the overall protection concept shall be agreed with affected parties and comply with the relevant standards concerning corrosion protection.

Examples of protection methods for endangered structures are given in Annex B.

NOTE Non-polarized (direct) drainage systems should not be connected to the traction system.

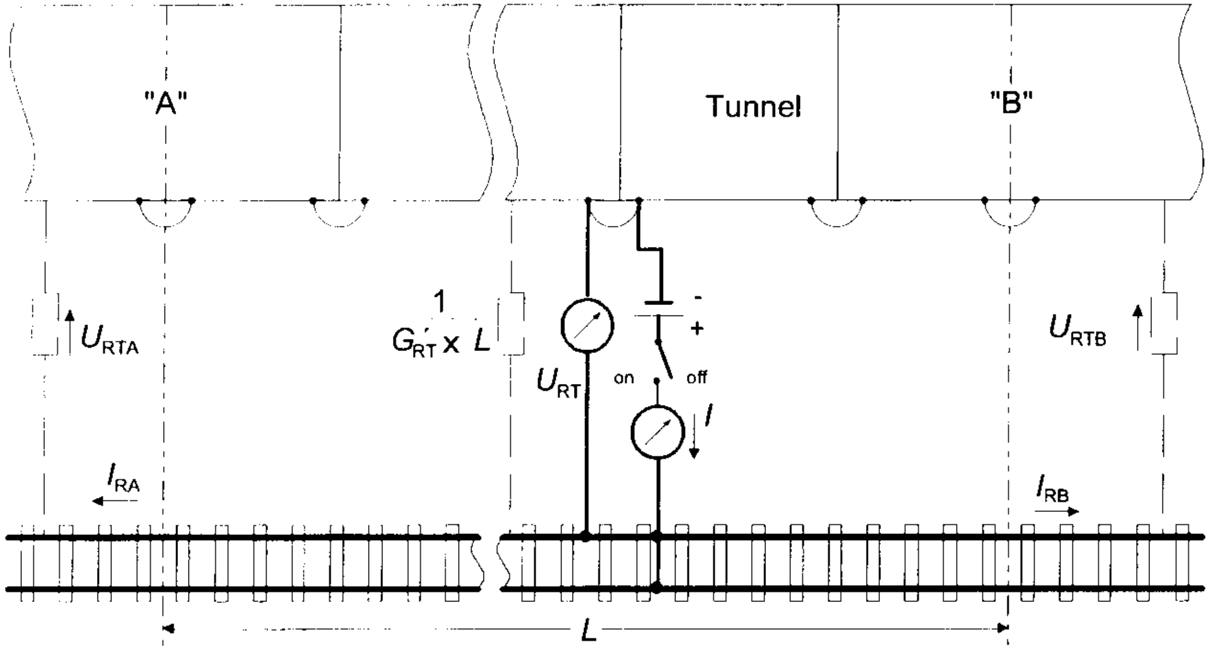


$$R_{R\ 10m} = \frac{(U_{A\ on} - U_{A\ off}) + (U_{B\ on} - U_{B\ off})}{I}$$

where

I	is the injected current, in Amperes;
$R_{R\ 10\ m}$	is the longitudinal resistance of a rail section of 10 m of rail 1 in Ohms: rail 2 accordingly;
$U_{on,\ off}$	is the voltage drop in rail 1 in Volts, with and without injected current; rail 2 accordingly.

Figure A.1 — Measurement of the rail resistance for a rail of 10 m length

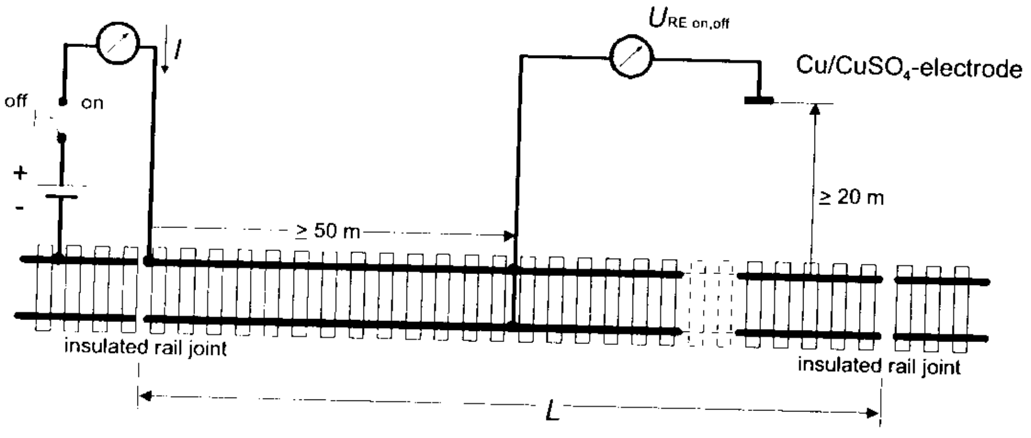


$$G'_{RT} = \frac{3}{L} \times \frac{I - I_{RA} - I_{RB}}{\Delta U_{RT} + \Delta U_{RTA} + \Delta U_{RTB}} \qquad \Delta U = U_{on} - U_{off}$$

where

G'_{RT}	is the conductance per unit length between rails and tunnel, in Siemens per kilometre;
I	is the injected current, in Amperes;
I_{RA}, I_{RB}	is the current flowing beyond the ends A, B of the measured section, in Amperes;
L	is the length of the section to be measured, in kilometres;
U_{RT}	is the voltage between the rail and the tunnel at the point of injection, in Volts;
U_{RTA}, U_{RTB}	is the voltage between the rail and the tunnel at the ends A, B of the tunnel section, in Volts.

Figure A.2 — Measuring arrangement for the conductance per unit length, G'_{RT}, between rails and tunnel



$$G'_{RE} = \frac{1}{L} \cdot \frac{I}{U_{REon} - U_{REoff}}$$

where

G'_{RE}	is the conductance per unit length between track and earth in Siemens per kilometre;
I	is the injected current, in Amperes;
L	is the length of the section to be measured, in kilometres;
U_{RE}	is the voltage between the rail and earth, in Volts.

Figure A.3 — Determination of the conductance per unit length G'_{RE} for track sections in open air

B.4 Impressed current cathodic protection

A system in which the impressed current is caused to flow from the impressed current anode to the protected structure via earth. An example is given in Figure B.3.

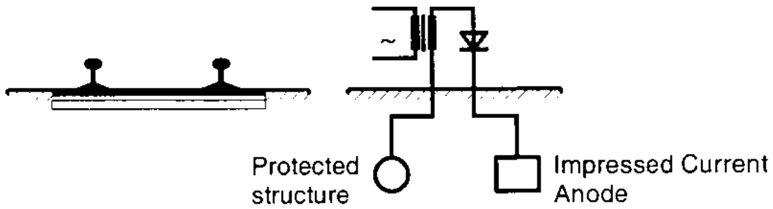


Figure B.3 — Impressed current cathodic protection

Annex C (informative) Estimation of the longitudinal voltage in reinforced railway structures

The longitudinal voltage in reinforced structures of tunnels, viaducts and reinforced track beds can be used for the assessment of stray current effects.

If the longitudinal voltage is less than 0,1 V, the conditions according to **7.2.3** are matched.

The longitudinal voltage in reinforced railway structures, caused by train operation, depends on the following affecting parameters:

— length of the considered line section;
— length of the adjacent line sections;
— conductance per unit length between the running rails and the structure;
— conductance per unit length between the structure and earth;
— longitudinal resistance of the running rails;
— longitudinal resistance of the interconnected structure;
— traction return current of the considered line section;
— traction return currents of adjacent lines.

For the calculation of the longitudinal voltage drop U_T of the interconnected structure for one line section equation (1) can be used.

NOTE The calculation method in equation (1) is very conservative. The formula assumes a infinitely long tunnel on each site of the considered section. Furthermore it doesn't take into account the reducing effects of the train movement in adjacent sections and the conductance per unit length of the tunnel structure versus earth. The calculated values can be much higher than in reality.

If the result of the calculation is higher than 0,1 V, a more detailed calculation method should be used.

$$U_T = 0{,}5 \times I \times L \times \frac{R'_R \times R'_T}{(R'_R + R'_T)} \times \left\{ 1 - \frac{L_C}{L} \times \left(1 - e^{-\left(\frac{L}{L_C}\right)} \right) \right\} \quad (1)$$

$$L_C = 1/\sqrt{(R'_R + R'_T) \times G'_{RT}} \quad (2)$$

where

G'_{RT}	is the conductance per unit length, in Siemens per kilometre;
I	is the average value of the traction return current of the considered section in the hour of the highest load, in Amperes;
L	is the length of the considered line section, in kilometres;
L_C	is the characteristic length of the system running rails/structure, in kilometres;
R'_R	is the resistance of the running rails per unit length, in Ohms per kilometre;
R'_T	is the resistance of the interconnected structure per unit length, in Ohms per kilometre;
U_T	is the longitudinal voltage in reinforced railway structure, in Volts.

Annex D (informative)
Bibliography

IEC 50(811), *International Electrotechnical Vocabulary — Chapter 811: Electric traction.*